CHAPTER 8

LOCOMOTION AND MOVEMENT

CHAPTER ESSENCE AT A GLANCE:
Sliding filament cross-bridge power strokes, troponin-calcium regulation, and articulation at synovial skeletal joints.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Sarcomere Microstructure and Myofilaments:** The sarcomere is the fundamental anatomical and functional unit of muscle contraction between two successive Z-lines: Thin filaments (F-actin polymers intertwined with two strands of tropomyosin, containing troponin complexes with calcium-binding subunits); Thick filaments (myosin polymers of meromyosin, featuring a globular Heavy Meromyosin HMM head with an actin-binding site and ATPase catalytic activity, and a Light Meromyosin LMM tail).

• **The Sliding Filament Mechanism:** Excitation-contraction coupling: Motor neuron action potential triggers acetylcholine release at the neuromuscular junction; muscle depolarization triggers sarcoplasmic reticulum to release Ca2+ into the sarcoplasm; Ca2+ binds to troponin C, unmasking myosin-binding sites on actin; myosin head hydrolyzes ATP, binds actin to form a cross-bridge, and performs a power stroke pulling actin toward the sarcomere center (H-zone disappears, I-band shortens, A-band remains constant).

• **Human Skeletal Organization and Rib Framework:** Adult skeleton comprises 206 bones: Axial skeleton (80 bones: skull 22, hyoid 1, ear ossicles 6, vertebral column 26, sternum 1, ribs 12 pairs); Appendicular skeleton (126 bones: pectoral girdle 4, upper limbs 60, pelvic girdle 2, lower limbs 60). Ribs: True ribs (pairs 1 to 7 attach directly to sternum via costal cartilages), Vertebrochondral false ribs (pairs 8 to 10 join 7th rib cartilage), and Floating ribs (pairs 11 and 12 unattached anteriorly).

• **Joint Classifications and Skeletal Pathology:** Joints: Fibrous joints (immovable sutures in skull), Cartilaginous joints (limited movement, intervertebral discs), and Synovial joints (freely movable with synovial fluid cavity: Ball and socket in shoulder or hip, Hinge in knee or elbow, Pivot in atlas-axis, Gliding between carpals, Saddle between carpal and metacarpal of thumb). Pathology: Myasthenia gravis (autoimmune neuromuscular junction disease), Tetany (low calcium spasms), Gout (uric acid crystals), and Osteoporosis (estrogen-deficient bone mass loss).

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\text{Cross-Bridge Cycle} = \text{Myosin-ATP} \xrightarrow{\text{Hydrolysis}} \text{Myosin-ADP-P}_i \xrightarrow{\text{Bind Actin}} \text{Cross-bridge} \longrightarrow \text{Power Stroke}$$

Actin-Myosin Cross-Bridge Power Stroke Cycle

$$\text{Sarcomere Band Dynamics} = \frac{\text{A-band: Constant length}}{\text{I-band: Shortens}} + \frac{\text{H-zone: Disappears}}{\text{Z-lines: Move inward}}$$

Sarcomeric Dimensional Changes during Active Contraction

$$\text{Rib Cage Taxonomy} = \text{True (1-7: Sternal)} + \text{False (8-10: 7th Cartilage)} + \text{Floating (11-12: Free anteriorly)}$$

Classification of Human Ribs by Sternal Articulation

$$\text{Synovial Archetypes} = \text{Ball-Socket (Shoulder)} + \text{Hinge (Elbow)} + \text{Pivot (Atlas-Axis)} + \text{Saddle (Thumb)}$$

Functional Typology of Freely Movable Synovial Joints

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** During muscle contraction, the length of the A-band NEVER changes; only the I-band shortens and the H-zone narrows or completely disappears.

• **Alert:** Myasthenia gravis is an autoimmune disorder attacking acetylcholine receptors, whereas Muscular dystrophy is an inherited genetic disorder.

• **Alert:** Gout is the inflammation of joints caused by accumulation of uric acid crystals, distinct from wear-and-tear osteoarthritis.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Describe the changes that occur in the A-band, I-band, and H-zone of a sarcomere during active muscle contraction. [Score: 2 Marks]

Answer & Key Valuation Steps:

• A-band: Retains its original constant length throughout contraction because thick myosin filaments do not change in length. **[0.75 Mark]**

• I-band: Shortens significantly as thin actin filaments slide into the A-band. **[0.75 Mark]**

• H-zone: Progressively narrows and completely disappears as opposing actin filaments overlap in the center. **[0.5 Mark]**

2. Distinguish between true ribs, false ribs, and floating ribs based on their anterior sternal attachments. [Score: 2 Marks]

Answer & Key Valuation Steps:

• True ribs (pairs 1 to 7): Attach directly to the sternum through hyaline costal cartilages. **[0.75 Mark]**

• False ribs (pairs 8, 9, 10): Do not articulate directly with the sternum; their cartilages join the costal cartilage of the 7th rib. **[0.75 Mark]**

• Floating ribs (pairs 11 and 12): Have no anterior attachment and remain free. **[0.5 Mark]**

3. Explain the sliding filament mechanism of muscle contraction from neuromuscular excitation to cross-bridge detachment, detailing the regulatory role of calcium ions and troponin. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Action potential arrival: Impulse reaches neuromuscular junction, releasing acetylcholine and generating an action potential along the sarcolemma into T-tubules. **[0.75 Mark]**

• Calcium release and unmasking: Sarcoplasmic reticulum releases Ca^{2+} ions, which bind to troponin C, causing a conformational change that shifts tropomyosin away and unmasks active myosin-binding sites on actin. **[0.75 Mark]**

• Cross-bridge power stroke: Myosin head hydrolyzes ATP, binds to the unmasked actin to form a cross-bridge, and tilts in a power stroke pulling actin filaments toward the M-line. **[0.75 Mark]**

• Detachment: Binding of a fresh ATP molecule to the myosin head breaks the cross-bridge, resetting the head for another cycle. **[0.75 Mark]**

4. Describe the molecular architecture of the thick (myosin) and thin (actin, tropomyosin, troponin) myofilaments. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Thick filament (Myosin): Polymer of meromyosins; each meromyosin has a globular head with cross-arm (Heavy Meromyosin HMM) containing actin-binding site and ATP-binding site, and a fibrous tail (Light Meromyosin LMM). **[1.5 Marks]**

• Thin filament: Two helically wound strands of filamentous F-actin (composed of G-actin monomers); two strands of tropomyosin run along F-actin grooves; complex troponin units distributed at regular intervals mask myosin-binding sites in the resting state. **[1.5 Marks]**

5. Classify synovial joints into four major types, state their degree of mobility, and provide an anatomical example for each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Ball and socket joint: Permits movement in all three planes (e.g., shoulder joint between humerus and glenoid cavity; hip joint between femur and acetabulum). **[0.75 Mark]**

• Hinge joint: Permits movement in a single angular plane like a door hinge (e.g., knee joint, elbow joint). **[0.75 Mark]**

• Pivot joint: Permits rotational movement around a single axis (e.g., joint between atlas and axis vertebrae). **[0.75 Mark]**

• Saddle joint: Joint with reciprocal concave and convex articular surfaces (e.g., between carpal and metacarpal of thumb). **[0.75 Mark]**

6. Differentiate between Myasthenia gravis, Tetany, and Gout based on their underlying pathological etiologies. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Myasthenia gravis: An autoimmune neuromuscular disorder where autoantibodies attack acetylcholine receptors on the motor end plate, causing fatigue, weakening, and paralysis of skeletal muscles. **[1 Mark]**

• Tetany: Characterized by rapid, painful muscular spasms resulting from hypocalcemia (abnormally low calcium ion concentration in body fluids). **[1 Mark]**

• Gout: Inflammation and acute swelling of joints caused by the chronic metabolic deposition of uric acid crystals in synovial membranes. **[1 Mark]**

7. After strenuous exercise, a runner experiences severe muscle cramps and stiffness where muscle fibers remain rigidly locked in a contracted state without relaxing. If intracellular ATP levels drop to zero while sarcoplasmic calcium remains elevated, explain why the muscle cannot relax, specifying the state of the myosin-actin cross-bridges. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• Biochemical cause of stiffness: Cross-bridge detachment requires the binding of a new ATP molecule to the myosin head. **[1.5 Marks]**

• State of cross-bridges: When intracellular ATP is completely depleted, the myosin head remains permanently locked to actin in a persistent cross-bridge complex (analogous to rigor mortis). **[1.5 Marks]**

• Calcium effect: Without ATP, the Ca2+ active transport pumps in the sarcoplasmic reticulum cannot pump calcium back into storage, leaving troponin unmasked and perpetuating the rigid contraction. **[1 Mark]**